Linear Regression with Heteroskedastic Errors

Siddhant Chaudhary and Aditya Bhaskara
Kahlert School of Computing, University of Utah

Problem setup

There are n sources. Source i provides t observations

$$x_i^{(j)} \sim D_i, \qquad y_i^{(j)} = \langle x_i^{(j)}, \beta \rangle + \eta_i^{(j)}, \qquad \eta_i^{(j)} \sim \mathcal{N}(0, \sigma_i^2).$$

The regression vector $\beta \in \mathbb{R}^d$ is shared. The variances σ_i^2 are arbitrary, different, and **unknown**.

The only quality guarantee is the subset-of-signals condition

$$\left|\{i \in [n] : \sigma_i \le 1\}\right| \ge m,$$

without knowing which sources form this subset.

Why standard regression is insufficient

- Weighted least squares requires the unknown variances.
- Existing comparable methods give error bounds that need not vanish as n increases.
- If the reliable sources were known, the oracle error would be $\Theta(\sqrt{d/(mt)})$.

The aim is therefore to obtain

$$\|\widehat{\beta} - \beta\|_2 = o(1)$$

using only the observations and the lower bound m.

Assumptions on the covariates

1. **Scaling:** $\|x_i^{(j)}\|_2 \le 1$.
2. **Well-spreadness:** a sampled $2d$-tuple is well-conditioned with probability at least 0.9.
3. Observations are independent across sources and replicates.

For the two-observation result, a leave-one-out version of well-conditioning is used. These assumptions hold with high probability for standard random designs such as vectors sampled from a sphere.

The regression-to-mean reduction

For a well-conditioned tuple $\tau = (i_1, \ldots, i_{2d})$, choose stable coefficients satisfying

$$e_r = \sum_{j=1}^{2d} \alpha_{j,r}(\tau) x_{i_j}.$$

Then

$$u_{\tau,r} := \sum_{j=1}^{2d} \alpha_{j,r}(\tau) y_{i_j} = \beta_r + \sum_{j=1}^{2d} \alpha_{j,r}(\tau) \eta_{i_j}.$$

Thus every $2d$-tuple produces one heteroskedastic estimate of coordinate β_r.

Technical obstacle: tuple estimates are dependent because overlapping tuples reuse the same source noise.

One observation per source

With $t = 1$, the algorithm constructs tuple estimates for each coordinate and applies iterative truncation.

1. **Enumerate stable tuples.** Collect all well-conditioned $2d$-tuples τ.
2. **Construct coordinate estimates.** Form $u_{\tau,r}$ for all such τ and $r \in [d]$.
3. **Iteratively truncate.** Repeatedly clip estimates around the current center at decreasing scales.
4. **Aggregate coordinates.** Return $\widehat{\beta} = (\widehat{\beta}_1, \ldots, \widehat{\beta}_d)$.

Guarantee for $t = 1$

Theorem 1

If $m \ge n^{1-1/(4d)}$, then with high probability

$$\|\widehat{\beta} - \beta\|_2 \le \bar{O}\left(\frac{d^{3/2} n^{2d-1/2}}{m^{2d}}\right).$$

The algorithm uses $O(n^{2d})$ tuple-scale work, up to factors depending on d.

Why the dependent estimates still concentrate

Let

$$N \asymp \binom{n}{2d}, \qquad M \asymp \binom{m}{2d}, \qquad \Gamma = \binom{n}{2d-1}.$$

Here N is the number of stable tuples, M counts tuples made entirely from reliable sources, and Γ controls tuple overlap.

A weak cancellation property, proved with Janson-type concentration, replaces the independence assumption used in standard heteroskedastic mean estimation. It gives coordinate error

$$O\left(\frac{\sqrt{\Gamma N \log N}}{M}\right),$$

which yields the stated regression bound after combining all d coordinates.

Interpretation

The reduction amplifies a hidden reliable subset: although the source labels are unknown, all-good tuples create $\binom{m}{2d}$ bounded-variance estimates of each coordinate.

Two observations per source

The second observation permits sample splitting: selection uses the first replicate, while estimation uses fresh noise from the second.

1. **Screen with the first replicate.** Guess a reliable $2d$-tuple and test the remaining sources against it.
2. **Retain a pruned set T.** At least $0.9m$ reliable sources survive, while very high-variance sources become sparse.
3. **Build fresh tuple estimates.** Using only second-replicate observations from T, construct $u_{\tau,r}$.
4. **Take the empirical mode.** For window width 1,

$$\widehat{\beta}_r = \arg\max_{u \in \mathbb{R}} \sum_{\tau \in \binom{T}{2d} \cap T_{\text{good}}} \mathbf{1}\{u_{\tau,r} \in [u-1, u+1]\}.$$

Guarantee for $t = 2$

Theorem 2

If $m \ge n^{5/6}$, then with probability at least 0.9,

$$\|\widehat{\beta} - \beta\|_2 \le \bar{O}\left(d^3 \left(\frac{n^5}{m^6}\right)^{1/3}\right).$$

The algorithm again uses $O(n^{2d})$ tuple-scale work, up to factors depending on d.

Why the modal estimator succeeds

Reliable tuples produce a density peak centered at β_r. The screening step controls the number of surviving sources at every variance scale, preventing noisier tuples from producing a competing peak.

The second replicate is essential to the proof: the noise used to choose T is independent of the noise later used to estimate β.

Takeaways and open questions

- These are the first sub-constant recovery guarantees for regression in this subset-of-signals model.
- Two observations improve the reliable-source threshold to $n^{5/6}$.
- Open: reduce the dependence on d, characterize the optimal threshold as a function of t, and obtain matching lower bounds.

Correspondence: urssidd@gmail.com